GATE ME 90 Days Challenge

Day 70/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Theory of Machines + Vibrations + Machine Design	**TOM, Vibrations & MD Revision**	**30 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on a full revision of Theory of Machines (Grashof's law, quick-return ratio, cam follower acceleration, interference, reverted gear trains, flywheels, gyroscopic couple, balancing), Vibrations (compound pendulum, Dunkerley's method, vibration isolation) and Machine Design (theories of failure, shaft design, bearing life and belt drives). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A four-bar chain has links of 40 mm, 90 mm, 70 mm and 100 mm. To obtain a crank–rocker mechanism, the link to be fixed is

(A) the 40 mm link

(B) the link opposite to the 40 mm link

(C) either link adjacent to the 40 mm link

(D) no link: the chain cannot rotate fully

Q2 NAT

In a crank and slotted-lever quick-return mechanism, the distance between the fixed centres is 300 mm and the crank is 150 mm long. The ratio of the time of the cutting stroke to the time of the return stroke is _____

Q3 NAT

A cam rotating at a uniform 600 rpm moves a knife-edge follower outward through 40 mm with simple harmonic motion during 90° of cam rotation. The maximum acceleration of the follower during the outstroke (in m/s², round off to one decimal place) is _____

Q4 MCQ

For 20° full-depth involute teeth (addendum = 1 module), the minimum number of teeth on a pinion meshing with a rack, to avoid interference, is

(A) 14

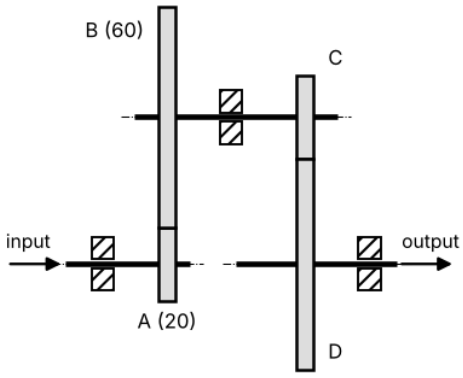
(B) 17

(C) 13

(D) 18

Q5 NAT

In the reverted gear train shown, the input and output shafts are coaxial and all the gears have the same module. Gear A (20 teeth) on the input shaft drives B (60 teeth) on the countershaft, and C on the countershaft drives D on the output shaft. If the overall speed reduction is 12, the number of teeth on D is _____



Q6 MCQ

A flywheel of given radius is designed for a certain fluctuation of energy and a given coefficient of fluctuation of speed. If the mean speed of the engine is doubled, the other quantities remaining the same, the mass of the flywheel required becomes

(A) twice the original
(B) one-half of the original
(C) one-quarter of the original
(D) four times the original

Q7 NAT

A disc of mass 5 kg and radius of gyration 100 mm spins at 1000 rpm about its axis, and the axis precesses at 2 rad/s in a plane perpendicular to the spin plane. The gyroscopic couple (in N·m, round off to two decimal places) is _____

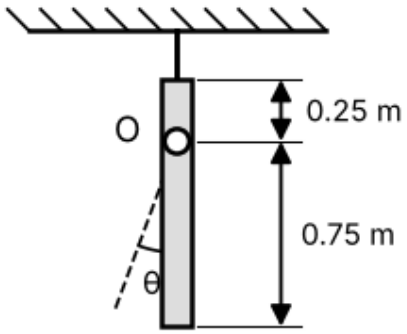
Q8 MCQ

A rigid rotor carries several unbalanced masses, all lying in a single plane perpendicular to the shaft axis. Which statement is correct?

(A) It needs balancing masses in two planes for dynamic balance
(B) If it is statically balanced ($\Sigma mr = 0$), it is also dynamically balanced
(C) Static balance is impossible to obtain for such a rotor
(D) Its unbalanced couple is $\Sigma mr \times$ (distance between the bearings)

Q9 NAT

A uniform rod 1 m long is pivoted about a horizontal axis through a point O, 0.25 m from its upper end, and swings in a vertical plane, as shown. Taking g = 9.81 m/s², the period of small oscillations (in s, round off to two decimal places) is _____



Q10 MCQ

Dunkerley's method for the fundamental natural frequency of a shaft carrying several discs gives

(A) the exact value

(B) an upper bound (a value higher than the true one)

(C) a lower bound (a value lower than the true one)

(D) the second natural frequency

Q11 NAT

A machine running at 1500 rpm is mounted on undamped isolators which deflect by 4 mm under its weight. Taking g = 9.81 m/s², the transmissibility (round off to three decimal places) is _____

Q12 MCQ

For a component made of a brittle material such as grey cast iron under a static two-dimensional stress system, the theory of failure normally used is the

(A) maximum principal stress (Rankine) theory

(B) maximum shear stress (Tresca) theory

(C) maximum distortion energy (von Mises) theory

(D) maximum strain energy (Haigh) theory

Q13 NAT

A solid shaft is to transmit 20 kW at 500 rpm. If the permissible shear stress is 40 MPa, the minimum shaft diameter required (in mm, round off to one decimal place) is _____

Q14 MCQ

The equivalent load on a deep-groove ball bearing is to be changed so that its rating life, at the same speed, is doubled. The load must be reduced to about

(A) 50 % of its original value

(B) 79 % of its original value

(C) 71 % of its original value

(D) 81 % of its original value

Q15 MCQ

For a belt drive in which centrifugal tension is considered, the power transmitted is maximum when the centrifugal tension equals

(A) one-half of the maximum tension

(B) one-third of the maximum tension

(C) the maximum tension

(D) one-quarter of the maximum tension

Answer Key

Q	Type	Answer
Q1	MCQ	(C) either link adjacent to the 40 mm link
Q2	NAT	2
Q3	NAT	315.8 (accept 313.0 to 318.5)
Q4	MCQ	(D) 18
Q5	NAT	64
Q6	MCQ	(C) one-quarter of the original
Q7	NAT	10.47 (accept 10.40 to 10.55)
Q8	MCQ	(B) If it is statically balanced ($\Sigma mr = 0$), it is also dynamically balanced
Q9	NAT	1.53 (accept 1.52 to 1.54)
Q10	MCQ	(C) a lower bound (a value lower than the true one)
Q11	NAT	0.110 (accept 0.109 to 0.112)
Q12	MCQ	(A) maximum principal stress (Rankine) theory
Q13	NAT	36.5 (accept 36.3 to 36.7)
Q14	MCQ	(B) 79 % of its original value
Q15	MCQ	(B) one-third of the maximum tension

Detailed Solutions

Q1 **Answer: (C) either link adjacent to the 40 mm link**

Concept: Grashof's law: if $s + l \le p + q$, at least one link can rotate fully.

$s + l = 40 + 100 = 140 < p + q = 90 + 70 = 160$: a Grashof chain.

Fixing a link adjacent to the shortest gives a crank–rocker; fixing the shortest gives a double crank; fixing the link opposite it gives a double rocker: **(C)**.

Exam tip: Note that the order of the links in the chain matters: 'adjacent' and 'opposite' refer to their positions, not their lengths.

Q2 **Answer: 2**

Concept: At the extreme positions the crank is perpendicular to the slotted lever, so $\cos(\beta/2) = r/c$, where β is the crank angle for the return stroke.

$\cos(\beta/2) = 150/300 = 0.5 \rightarrow \beta/2 = 60°$, return angle $\beta = 120°$

Cutting angle $= 360° - 120° = 240°$; ratio $= 240/120 =$ **2**

Exam tip: A longer crank (relative to the centre distance) gives a larger quick-return ratio.

Q3 **Answer: 315.8 (accept 313.0 to 318.5)**

Concept: For SHM over a cam angle θ_o, $a_{max} = (h/2)(\pi\omega/\theta_o)^2$.

$\omega = 2\pi \times 600/60 = 62.83$ rad/s; $\theta_o = \pi/2$ rad, so $\pi\omega/\theta_o = 2\omega = 125.66$ rad/s

$a_{max} = 0.02 \times 125.66^2 =$ **315.8 m/s²**

Exam tip: Halving the cam angle for the same lift multiplies the peak acceleration by four.

Q4 **Answer: (D) 18**

Concept: For a pinion meshing with a rack, interference is avoided if $T \geq 2a_w/\sin^2\varphi$ (a_w = addendum in modules).

$T_{min} = 2 \times 1/\sin^2 20° = 2/0.11698 = 17.1$

Rounding up: **18 teeth (D)**.

Exam tip: 17 is too few (17 < 17.1); always round up. For 14.5° teeth the minimum is 32.

Q5 **Answer: 64**

Concept: Speed ratio = product of (driven/driver) teeth ratios; coaxial shafts with the same module need equal centre distances: $T_A + T_B = T_C + T_D$.

$(60/20)(T_D/T_C) = 12 \rightarrow T_D = 4T_C$

$T_C + T_D = 20 + 60 = 80 \rightarrow 5T_C = 80 \rightarrow T_C = 16$, $T_D =$ **64**

Exam tip: Reverted trains (lathe back gears, gearbox lay shafts) always need this centre-distance condition.

Q6 **Answer: (C) one-quarter of the original**

Concept: $\Delta E = I\omega^2 C_s = mk^2\omega^2 C_s$.

For the same ΔE, C_s and k: $m \propto 1/\omega^2$

Doubling the speed makes the required mass **one-quarter (C)**.

Exam tip: This is why high-speed engines need only small flywheels.

Q7 **Answer: 10.47 (accept 10.40 to 10.55)**

Concept: Gyroscopic couple $C = I\omega\omega_p$.

$I = mk^2 = 5 \times 0.1^2 = 0.05$ kg·m²; $\omega = 2\pi \times 1000/60 = 104.72$ rad/s

$C = 0.05 \times 104.72 \times 2 =$ **10.47 N·m**

Exam tip: Use the spin speed in rad/s; rpm in the formula is the most common error.

Q8 **Answer: (B) If it is statically balanced (Σmr = 0), it is also dynamically balanced**

Concept: Dynamic balance needs $\Sigma mr = 0$ (no resultant force) and $\Sigma mrl = 0$ (no resultant couple).

With all masses in one plane, l is the same for every mass, so $\Sigma mrl = l\,\Sigma mr$: static balance automatically removes the couple: **(B)**.

Masses in several planes need balancing in two planes.

Exam tip: Thin discs and pulleys are balanced statically; long rotors (motor armatures, crankshafts) need dynamic balancing.

Q9 **Answer: 1.53 (accept 1.52 to 1.54)**

Concept: Compound pendulum: $T = 2\pi\sqrt{(I_O/(mgd))}$, d = distance from the pivot to the centre of mass.

$d = 0.5 - 0.25 = 0.25$ m; $I_O = mL^2/12 + md^2 = m(0.0833 + 0.0625) = 0.1458m$

$T = 2\pi\sqrt{(0.1458/(9.81 \times 0.25))} = 2\pi \times 0.2438 =$ **1.53 s**

Exam tip: The equivalent simple-pendulum length is $I_O/(md) = 0.583$ m, longer than d.

Q10 **Answer: (C) a lower bound (a value lower than the true one)**

Concept: Dunkerley: $1/f^2 = 1/f_0^2 + 1/f_1^2 + 1/f_2^2 + \ldots$ (shaft alone and each disc alone).

Neglecting the cross terms makes the right side too large, so f is underestimated: **a lower bound (C)**.

Rayleigh's energy method, by contrast, gives an upper bound.

Exam tip: The true fundamental frequency lies between the Dunkerley and Rayleigh estimates.

Q11 **Answer: 0.110 (accept 0.109 to 0.112)**

Concept: $\omega n = \sqrt{(g/\delta)}$; for an undamped mount $TR = 1/(r^2 - 1)$ when $r > 1$.

$\omega n = \sqrt{(9.81/0.004)} = 49.52$ rad/s; $\omega = 2\pi \times 1500/60 = 157.08$ rad/s; $r = 3.172$

$TR = 1/(3.172^2 - 1) = 1/9.06 =$ **0.110** (89 % isolation)

Exam tip: Softer isolators (larger static deflection) lower ωn and improve isolation.

Q12 **Answer: (A) maximum principal stress (Rankine) theory**

Concept: Brittle materials fail by fracture on the plane of maximum tensile stress, without yielding.

Hence the **maximum principal stress theory (A)** (or the Coulomb–Mohr theory) is used, with S_ut as the limit.

Tresca and von Mises predict yielding of ductile materials.

Exam tip: von Mises gives shear yield strength = 0.577 S_y; Tresca gives 0.5 S_y.

Q13 **Answer: 36.5 (accept 36.3 to 36.7)**

Concept: $T = 60P/(2\pi N)$ and $\tau = 16T/(\pi d^3)$.

$T = 60 \times 20\,000/(2\pi \times 500) = 381.97$ N·m

$d = (16 \times 381\,970/(\pi \times 40))^{1/3} = 48\,634^{1/3} =$ **36.5 mm**

Exam tip: In practice the next standard size (38 or 40 mm) is chosen.

Q14 **Answer: (B) 79 % of its original value**

Concept: For ball bearings $L \propto (C/P)^3$, so $P \propto L^{-1/3}$.

$P_{new}/P_{old} = 2^{-1/3} = 0.794$

The load must fall to about **79 % (B)**; for a roller bearing (exponent 10/3) it would be 81 %.

Exam tip: Bearing life is very sensitive to load: a 10 % overload cuts ball-bearing life by about 25 %.

Q15 **Answer: (B) one-third of the maximum tension**

Concept: $P = (T_1 - T_2)v = (T_{max} - mv^2)(1 - e^{-\mu\theta})v$.

$dP/dv = 0 \rightarrow T_{max} - 3mv^2 = 0$, so $T_c = mv^2 = T_{max}/3$: **(B)**

The optimum belt speed is $v = \sqrt{(T_{max}/(3m))}$.

Exam tip: Centrifugal tension does not change the initial-tension balance; it only reduces the tension available for power.

What next?

Day 70 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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